

Project Title: Design and Development of a Smart Hospital Management System: A Case Study of **MediCare Hospital**.

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2. Declaration

Norman Odhiambo, Yemi Justus, Dickson Japhet and Faraday Mayaka do hereby declare that this project proposal, titled "Design and Development of a Smart Hospital Management System: A Case Study of **MediCare Hospital**" is our own original work. It has been prepared for the partial fulfillment of the requirements for the Bachelor of Science in Computer Science degree. All sources of information used in this proposal have been duly acknowledged.

3. Acknowledgement

We wish to express our sincere gratitude to my supervisor, Mr. Arnold, for their invaluable guidance, encouragement, and constructive feedback throughout the initial stages of this project.

I would also like to thank the administration and staff of **MediCare Hospital** for their cooperation and for providing us with the necessary insights into their current operations, which formed the basis of this study.

4. Abstract

This proposal outlines the development of a Smart Hospital Management System (SHMS) for **MediCare Hospital**. The current manual or semi-digital methods for patient registration, appointment booking, and doctor-patient communication are inefficient, leading to long queues, administrative bottlenecks, and poor patient experience. This project aims to design and implement a centralized, web-based system to automate these core processes. The proposed system will feature distinct portals for patients, doctors, and administrators. Patients will be able to book appointments online, view doctor feedback on their dashboard, and receive email notifications. Doctors will have a dedicated interface to manage their schedules and respond to patient appointments. An admin panel will provide complete oversight, allowing for the management of all users, doctors, and appointments. The system will be developed using an agile methodology to ensure flexibility and user-centered design. The successful implementation of this SHMS is expected to significantly enhance operational efficiency, improve patient satisfaction, and provide a secure, centralized repository for hospital management data.

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6. Study Background

This project focuses on the research area of **Healthcare Information Systems**, specifically the application of web-based technologies to improve hospital administration and patient care. The case study for this project is **MediCare Hospital**, a medium-sized community hospital located in Rusinga Island. The hospital offers services in various departments including general medicine, pediatrics, and cardiology.

Currently, **MediCare Hospital** relies on a largely manual system for managing patient flow. Patients must physically visit the hospital or call the reception desk to book appointments. Appointment details are recorded in paper-based ledgers or disparate spreadsheets. Doctor feedback and prescriptions are handwritten and given to the patient, with no centralized digital record. Communication between doctors and patients regarding follow-ups or test results is done via phone calls, which is time-consuming and lacks a paper trail. The hospital administration struggles to generate reports, track doctor performance, or manage patient data efficiently due to the fragmented nature of the current system.

7. Problem Statement

The existing operational methods at **MediCare Hospital** present several critical challenges:

- **Inefficient Appointment Booking:** Patients face difficulties in booking appointments, often experiencing long wait times on the phone or having to make unnecessary trips to the hospital. Double-booking and scheduling conflicts are common.
- **Poor Doctor-Patient Communication:** There is no structured, trackable method for doctors to provide feedback, test results, or prescriptions to patients. This can lead to miscommunication and lost information.

- **Fragmented Data Management:** Patient medical histories, doctor schedules, and appointment records are scattered across paper files and spreadsheets. This makes it difficult to retrieve information quickly, compromising the quality of care and administrative efficiency.
 - **Lack of Centralized Control:** The hospital administration has no single, unified interface to oversee operations. Managing doctor profiles, patient records, or canceling/rescheduling appointments is a cumbersome manual task. The inability to easily delete or update records poses data integrity and security risks.
 - **Poor Patient Experience:** The combination of these factors leads to a frustrating patient experience, characterized by long wait times, administrative confusion, and a lack of transparent communication.
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8. Proposed Solution

This project proposes the development of a **web-based Smart Hospital Management System (SHMS)** tailored for **MediCare Hospital**. This system will serve as a centralized digital platform to automate and streamline the hospital's core administrative and clinical workflows.

The SHMS will feature three primary interfaces:

1. **Patient Portal:** Allows patients to register, book appointments with specific doctors, view the status and feedback from their doctor, and receive automated email confirmations and updates.
2. **Doctor Portal:** Enables doctors to view their daily schedules, access patient appointment details, and provide diagnostic feedback, prescriptions, and notes, which are instantly saved and shared.
3. **Admin Portal:** Provides the administrator with full control over the system, including the ability to add, update, or delete doctor profiles, manage patient records, and oversee, reschedule, or cancel all appointments.

Justification: This solution is justified as it directly addresses the identified problems. By digitizing these processes, the SHMS will eliminate inefficiencies, enhance communication, ensure data integrity, and provide a robust, scalable platform for the hospital's future growth. The use of distinct roles ensures data security and appropriate access levels.

9. Project Goal and Objectives

Project Goal:

To improve the operational efficiency, data management, and patient experience at **MediCare Hospital** by designing and implementing a secure and user-friendly Smart Hospital Management System.

Objectives:

To achieve this goal, the project will pursue the following objectives:

1. To conduct a comprehensive requirement analysis by studying the current workflows and information needs of administrators, doctors, and patients at the case study hospital.
2. To design a database schema that securely and efficiently stores user profiles, doctor details, patient records, and appointment information.
3. To develop a responsive web application with separate modules for patients, doctors, and administrators, ensuring role-based access control.
4. To implement a booking engine that allows patients to view doctor availability and schedule, reschedule, or cancel appointments.
5. To integrate an automated email notification system to confirm appointments and relay doctor feedback to patients.
6. To develop a comprehensive admin dashboard that allows for complete management (CRUD) of all system entities (doctors, patients, appointments).
7. To test the system thoroughly with end-users from the case study hospital to ensure it meets their needs and is free of critical bugs.

10. Research Questions

To ensure the proposed system is built on a foundation of current knowledge and best practices, the literature review will focus on the following research questions:

1. **State of the Art:** What are the current trends and technologies used in modern web-based hospital management systems, particularly concerning patient portals and electronic health records (EHRs)?
2. **User Experience (UX):** What are the best practices in designing user interfaces for different user groups (patients vs. medical professionals) in healthcare applications to ensure usability and adoption?

3. **Security and Privacy:** What are the standard security protocols and data protection regulations (e.g., data protection laws) that must be considered when developing a system that handles sensitive patient data?
 4. **Communication Efficacy:** How can integrated communication tools (such as in-app dashboards and automated emails) improve the effectiveness of doctor-patient communication compared to traditional methods?
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11. System Development Methodology

This project will be developed using the **Agile Scrum methodology**. This framework is chosen because it is iterative, flexible, and highly collaborative, which is ideal for a project where requirements may evolve based on user feedback.

Justification: Agile Scrum allows for the development of the system in small, manageable increments called "sprints." This approach ensures that a potentially usable version of the software is available early in the process, and feedback from the client (the hospital) can be incorporated into subsequent sprints. This reduces the risk of developing a system that does not meet the actual needs of its users.

Phases of the Methodology:

1. **Project Initiation & Product Backlog Creation:** Define the overall project vision and create a prioritized list of all desired features (user stories) for the system.
2. **Sprint Planning:** Select a set of high-priority user stories from the product backlog to be completed in a 2-week sprint. This first sprint might focus on user registration and login.
3. **Sprint Execution (Design & Development):**
 - **Design:** Create wireframes for the user interfaces and design the database schema.
 - **Development:** Write the code for the features planned in the sprint. The tech stack will likely include **HTML, CSS, JavaScript** for the frontend, **Node.js/Express** for the backend, and **MySQL** for the database.
4. **Sprint Review & Retrospective:** Demonstrate the completed features to the supervisor/client and gather feedback. Reflect on the sprint process to identify areas for improvement.

5. **Repeat:** The process repeats for subsequent sprints, adding features like the appointment booking engine, doctor response module, email integration, and the admin panel.
 6. **Testing & Deployment:** Conduct system-wide testing, including unit testing, integration testing, and user acceptance testing (UAT) with hospital staff, before final deployment.
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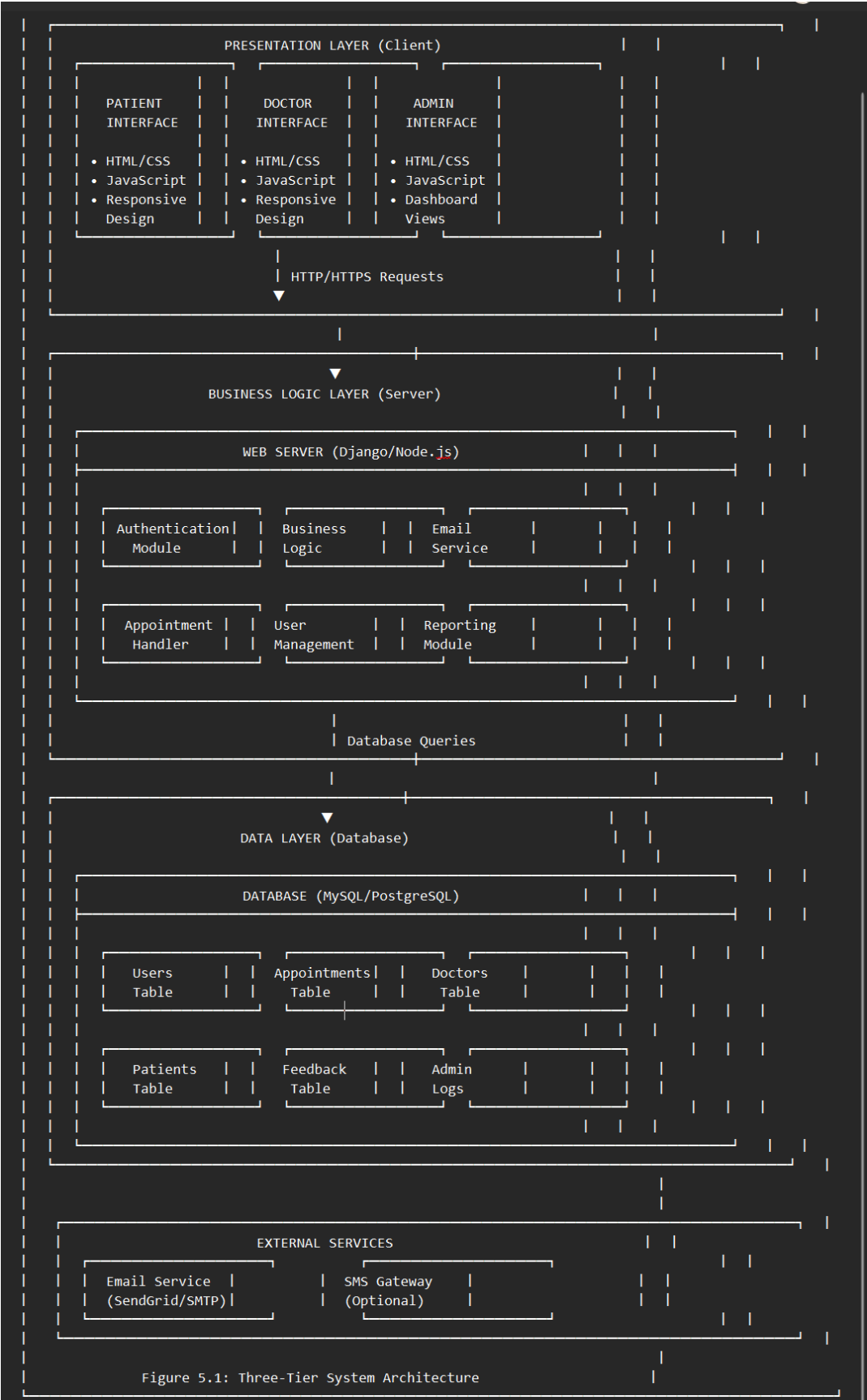
Figure 1.1 shows the Use Case Diagram for the Smart Hospital Management System, illustrating the interactions between the three actors (Patient, Doctor, Admin) and the system's core functionalities.



Figure 3.1: Use Case Diagram for SHMS

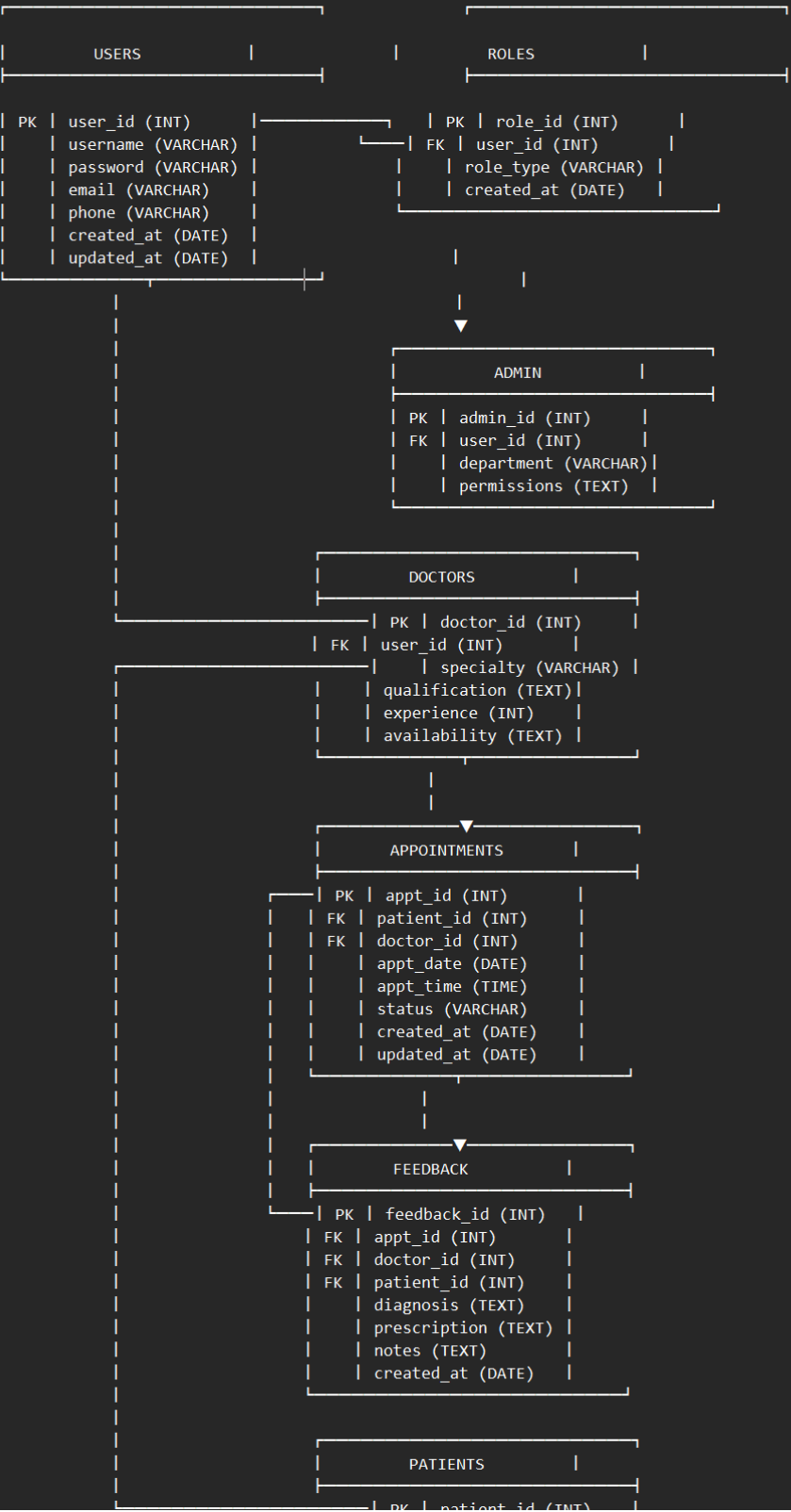
SYSTEM ARCHITECTURE DIAGRAM

Figure 1.2 illustrates the three-tier architecture of the Smart Hospital Management System, showing the separation between presentation layer, business logic layer, and data layer.



ENTITY RELATIONSHIP DIAGRAM (ERD)

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- **Development Laptop/PC:** A personal computer with at least 8GB RAM and a modern processor for coding, testing, and running the local server.
- **Smartphone/Tablet:** For testing the responsiveness of the web application on mobile devices.
- **External Hard Drive / Cloud Storage:** For regular backups of the project code and database.

Software:

- **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu).
- **Code Editor/IDE:** VS Code
- **Web Browser:** Google Chrome or Mozilla Firefox with developer tools.
- **Version Control:** Git and a GitHub account for code management and collaboration.
- **Backend Framework & Language:** JavaScript with Node.js/Express.
- **Frontend Technologies:** HTML5, CSS3, JavaScript (potentially a framework like React or Vue.js).
- **Database Management System:** MySQL.
- **Email Testing Tool:** Nodemailer for safe email testing during development.
- **Design & Prototyping:** Figma XD for UI/UX design.
- **Project Management:** Trello or Jira for managing the Scrum backlog.
- **Office Suite:** Microsoft Office or Google Workspace for documentation.

Other Resources:

- Access to the internet for research and troubleshooting.
- Access to **MediCare Hospital** personnel for requirement gathering and testing.

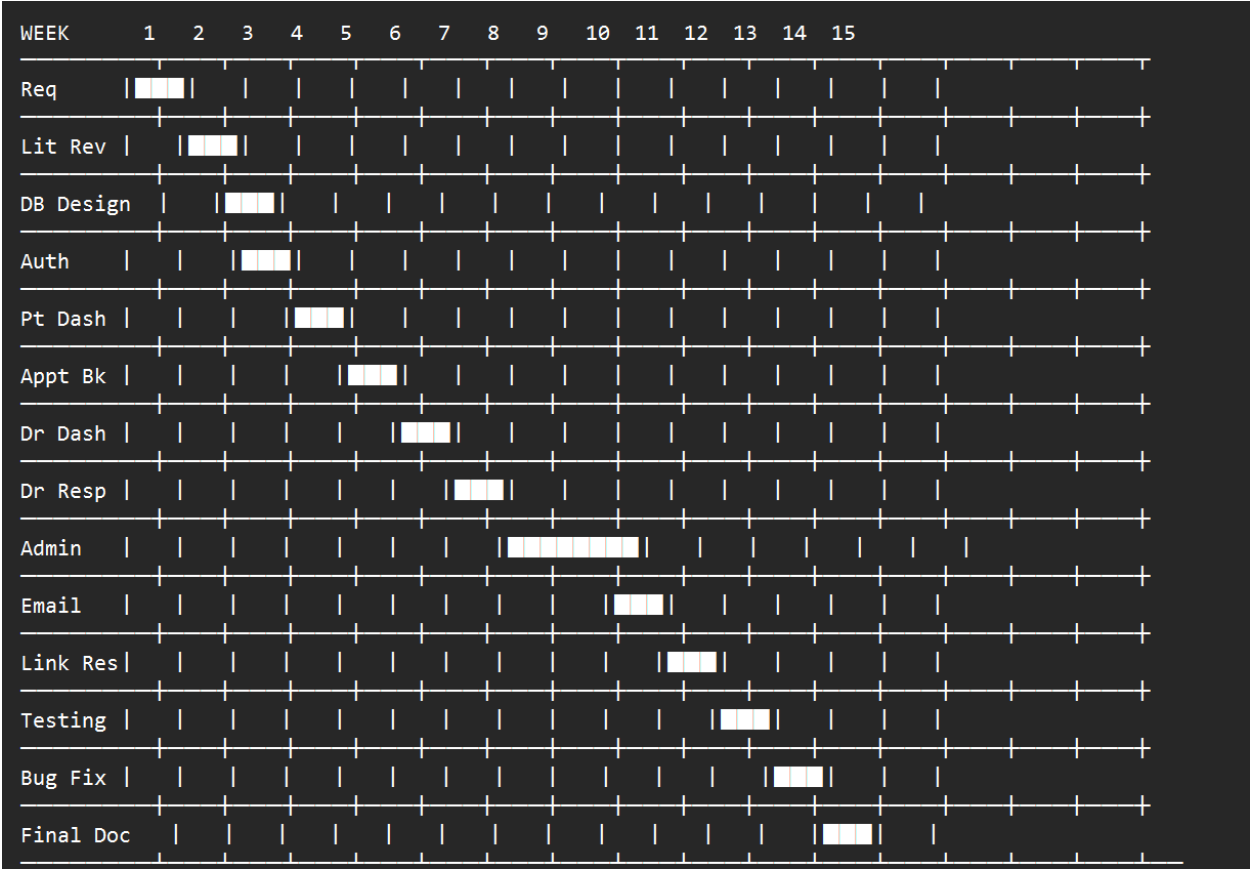
13. Project Schedule

The project is estimated to be completed over a period of 15 weeks, divided into 2-week sprints.

ID	Activity/Task	Start Date	End Date	Duration (Weeks)
1	Project Initiation & Research	Week 1	Week 2	2
2	Requirements Gathering & Analysis	Week 1	Week 1	1
3	Literature Review & Tech Stack Finalization	Week 2	Week 2	1
4	Sprint 1: Core Setup	Week 3	Week 4	2
5	Database Design & Setup	Week 3	Week 3	1
6	User Authentication (Registration/Login)	Week 4	Week 4	1
7	Sprint 2: Patient Module	Week 5	Week 6	2
8	Patient Dashboard & Profile	Week 5	Week 5	1
9	Appointment Booking Interface	Week 6	Week 6	1
10	Sprint 3: Doctor Module	Week 7	Week 8	2
11	Doctor Dashboard & Schedule View	Week 7	Week 7	1
12	Doctor Response/Prescription Feature	Week 8	Week 8	1
13	Sprint 4: Admin Module	Week 9	Week 10	2

ID	Activity/Task	Start Date	End Date	Duration (Weeks)
14	Admin Dashboard & User Management	Week 9	Week 10	2
15	Sprint 5: Integration & Features	Week 11	Week 12	2
16	Email Notification Integration	Week 11	Week 11	1
17	Linking Doctor Responses to Patient Dashboard	Week 12	Week 12	1
18	Testing & Deployment	Week 13	Week 14	2
19	System, Integration, and User Acceptance Testing	Week 13	Week 13	1
20	Bug Fixing & Final Deployment	Week 14	Week 14	1
21	Documentation & Submission	Week 15	Week 15	1

Gantt Chart:



14. Budget

As this is an academic project, most of the resources are already available. The budget is therefore minimal and primarily covers potential costs for deployment and final documentation.

Item	Description	Estimated Cost (Local Currency)
Software & Services		
Domain Name	For hosting the live project (e.g., .com) for one year	\$10 - \$15
Web Hosting	Shared hosting or a small VPS for 6 months	\$30 - \$50

Item	Description	Estimated Cost (Local Currency)
Email Service (Transactional)	If free tiers are exhausted (e.g., SendGrid)	\$0 - \$20
Consumables		
Printing & Binding	For final project report and proposal	\$15 - \$25
Stationery	Pens, notebooks for meetings and notes	\$5 - \$10
Contingency	For unforeseen expenses	\$10 - \$20
Total Estimated Cost		\$70 - \$140

15. Conclusion

The current manual systems at **MediCare Hospital** are a significant bottleneck to its growth and a detriment to patient satisfaction. This project proposal has outlined a clear, feasible, and impactful solution: the development of a Smart Hospital Management System. By automating key processes like appointment booking and doctor-patient communication, and by providing a powerful centralized control panel for administrators, this system will directly tackle the core operational problems. The successful implementation of this project will not only fulfill an academic requirement but will also deliver a tangible, valuable tool to the case study hospital, modernizing its operations and significantly enhancing the quality of service it can provide to its patients. Your support for this undertaking is therefore earnestly solicited.

16. References

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